

PRACTICAL 3

Task 1: Document Modelling

Document Modelling is the process of designing how data is stored inside MongoDB documents. Unlike SQL databases, MongoDB uses a flexible schema, so documents in the same collection can have different fields and structures.

- Created a `book_metadata` collection in MongoDB.
- Added details like title, author, genre, year, formats, rating, and specifications.

Used embedded documents for specs.

- Stored reviews inside each book document.
- Used `member_id` to reference members from the SQL database.
- Different books have different specifications, showing MongoDB's flexible structure.
- Inserted 4 book documents into the collection.

```
> use("BookFlowDB")
< switched to db BookFlowDB
> db.createCollection("book_metadata")
< { ok: 1 }
> show collections
< book_metadata
> db.book_metadata.insertOne({
  title: "The Silent Algorithm",
  author: "R. Menon",
  genre: "Technology",
  publishedYear: 2022,
  formats: ["Hardcover", "Digital", "Audiobook"],
  specs: {
    pages: 340,
    fileSize_MB: 4.2,
    drm: true,
    narrator: "K. Rao",
    runtime_min: 610
  },
  avgRating: 4.6,
  reviews: [
    {
      member_id: "M1001",
      rating: 5,
      comment: "Excellent deep dive into digital systems design.",
      date: ISODate("2023-01-15")
    },
  ],
})
```

```

    {
      member_id: "M1042",
      rating: 4,
      comment: "Good but the Digital edition had formatting issues.",
      date: ISODate("2023-02-20")
    }
  ]
})
< {
  acknowledged: true,
  insertedId: ObjectId('6a8e5d327171bd7b6f7ad40c')
}
> db.book_metadata.find().pretty()
< {
  _id: ObjectId('6a8d3d05400456cf7225be94'),
  title: 'The Silent Algorithm',
  author: 'R. Menon',
  genre: 'Technology',
  publishedYear: 2022,
  formats: [
    'Hardcover',
    'Digital',
    'Audiobook'
  ],

```

```

  specs: {
    pages: 340,
    fileSize_MB: 4.2,
    drm: true,
    narrator: 'K. Rao',
    runtime_min: 610
  },
  avgRating: 4.6,
  reviews: [
    {
      member_id: 'M1001',
      rating: 5,
      comment: 'Excellent deep dive into digital systems design.',
      date: 2023-01-15T00:00:00.000Z
    },
    {
      member_id: 'M1042',
      rating: 4,
      comment: 'Good but the Digital edition had formatting issues.',
      date: 2023-02-20T00:00:00.000Z
    }
  ]
}

```

```
{
  _id: ObjectId('6a8e5d327171bd7b6f7ad40c'),
  title: 'The Silent Algorithm',
  author: 'R. Menon',
  genre: 'Technology',
  publishedYear: 2022,
  formats: [
    'Hardcover',
    'Digital',
    'Audiobook'
  ],
```

```
  specs: {
    pages: 340,
    fileSize_MB: 4.2,
    drm: true,
    narrator: 'K. Rao',
    runtime_min: 610
  },
  avgRating: 4.6,
  reviews: [
    {
      member_id: 'M1001',
      rating: 5,
      comment: 'Excellent deep dive into digital systems design.',
      date: 2023-01-15T00:00:00.000Z
    },
    {
      member_id: 'M1042',
      rating: 4,
      comment: 'Good but the Digital edition had formatting issues.',
      date: 2023-02-20T00:00:00.000Z
    }
  ]
}
```

BookFlowDB > |

Task 2:

Advanced Querying

Using MongoDB operators to search and filter the books. Your practical specifically uses \$gt, \$in, and \$regex.

- Used **\$gt** to find books with rating greater than 4.
- Used **\$in** to find books available in Digital or Audiobook formats.
- Combined conditions to filter books based on **rating and format**.
- Used **\$regex** to search for the word **Digital**.
- Used dot notation to search **inside embedded review comments**.
- These queries help BookFlow quickly filter and search book information.

```
> db.book_metadata.find({ avgRating: {$gt: 4} });
< {
  _id: ObjectId('6a8d3d05400456cf7225be94'),
  title: 'The Silent Algorithm',
  author: 'R. Menon',
  genre: 'Technology',
  publishedYear: 2022,
  formats: [
    'Hardcover',
    'Digital',
    'Audiobook'
  ],
  specs: {
    pages: 340,
    fileSize_MB: 4.2,
    drm: true,
    narrator: 'K. Rao',
    runtime_min: 610
  },
  avgRating: 4.6,
  reviews: [
    {
      member_id: 'M1001',
      rating: 5,
      comment: 'Excellent deep dive into digital systems design.',
      date: 2023-01-15T00:00:00.000Z
    },
  ],
}
```

```
{
  member_id: 'M1042',
  rating: 4,
  comment: 'Good but the Digital edition had formatting issues.',
  date: 2023-02-20T00:00:00.000Z
}
]
}
{
  _id: ObjectId('6a8e5d327171bd7b6f7ad40c'),
  title: 'The Silent Algorithm',
  author: 'R. Menon',
  genre: 'Technology',
  publishedYear: 2022,
  formats: [
    'Hardcover',
    'Digital',
    'Audiobook'
  ],
  specs: {
    pages: 340,
    fileSize_MB: 4.2,
    drm: true,
    narrator: 'K. Rao',
    runtime_min: 610
  },
}
```

```
    avgRating: 4.6,
    reviews: [
      {
        member_id: 'M1001',
        rating: 5,
        comment: 'Excellent deep dive into digital systems design.',
        date: 2023-01-15T00:00:00.000Z
      },
      {
        member_id: 'M1042',
        rating: 4,
        comment: 'Good but the Digital edition had formatting issues.',
        date: 2023-02-20T00:00:00.000Z
      }
    ]
  }
}
> db.book_metadata.find(
  { formats: { $in: ["Digital", "Audiobook"] } },
  { title: 1, formats: 1, _id: 0 }
)
< {
  title: 'The Silent Algorithm',
  formats: [
    'Hardcover',
    'Digital',
    'Audiobook'
  ]
}
```

```
{
  title: 'The Silent Algorithm',
  formats: [
    'Hardcover',
    'Digital',
    'Audiobook'
  ]
}
> db.book_metadata.find({
  avgRating: { $gt: 4 },
  formats: { $in: ["Digital"] }
})
< {
  _id: ObjectId('6a8d3d05400456cf7225be94'),
  title: 'The Silent Algorithm',
  author: 'R. Menon',
  genre: 'Technology',
  publishedYear: 2022,
  formats: [
    'Hardcover',
    'Digital',
    'Audiobook'
  ],
```

```
specs: {
  pages: 340,
  fileSize_MB: 4.2,
  drm: true,
  narrator: 'K. Rao',
  runtime_min: 610
},
avgRating: 4.6,
reviews: [
  {
    member_id: 'M1001',
    rating: 5,
    comment: 'Excellent deep dive into digital systems design.',
    date: 2023-01-15T00:00:00.000Z
  },
  {
    member_id: 'M1042',
    rating: 4,
    comment: 'Good but the Digital edition had formatting issues.',
    date: 2023-02-20T00:00:00.000Z
  }
]
}
```



```
{
  _id: ObjectId('6a8e5d327171bd7b6f7ad40c'),
  title: 'The Silent Algorithm',
  author: 'R. Menon',
  genre: 'Technology',
  publishedYear: 2022,
  formats: [
    'Hardcover',
    'Digital',
    'Audiobook'
  ],
  specs: {
    pages: 340,
    fileSize_MB: 4.2,
    drm: true,
    narrator: 'K. Rao',
    runtime_min: 610
  },
  avgRating: 4.6,
  reviews: [
    {
      member_id: 'M1001',
      rating: 5,
      comment: 'Excellent deep dive into digital systems design.',
      date: 2023-01-15T00:00:00.000Z
    },
  ],
}
```

```
{
  member_id: 'M1042',
  rating: 4,
  comment: 'Good but the Digital edition had formatting issues.',
  date: 2023-02-20T00:00:00.000Z
}
]
}
> db.book_metadata.find(
  { formats: { $regex: "Digital", $options: "i" } },
  { title: 1, formats: 1, _id: 0 }
)
< {
  title: 'The Silent Algorithm',
  formats: [
    'Hardcover',
    'Digital',
    'Audiobook'
  ]
}
{
  title: 'The Silent Algorithm',
  formats: [
    'Hardcover',
    'Digital',
    'Audiobook'
  ]
}
```

```

> db.book_metadata.find(
  { "reviews.comment": { $regex: "Digital", $options: "i" } },
  { title: 1, "reviews.$": 1, _id: 0 }
)
< {
  title: 'The Silent Algorithm',
  reviews: [
    {
      member_id: 'M1001',
      rating: 5,
      comment: 'Excellent deep dive into digital systems design.',
      date: 2023-01-15T00:00:00.000Z
    }
  ]
}
{
  title: 'The Silent Algorithm',
  reviews: [
    {
      member_id: 'M1001',
      rating: 5,
      comment: 'Excellent deep dive into digital systems design.',
      date: 2023-01-15T00:00:00.000Z
    }
  ]
}

```

Task 3: Aggregation Pipeline

- Used an aggregation pipeline to generate a Top Rated report.
- \$match selected books published after 2020.
- \$unwind separated the reviews.
- \$group calculated the average rating and review count.
- \$sort arranged books by rating from highest to lowest.
- \$project displayed the required final fields.

```

> db.book_metadata.aggregate([
  { $match: { publishedYear: { $gt: 2020 } } },
  { $unwind: "$reviews" },
  {
    $group: {
      _id: "$title",
      avgReviewRating: { $avg: "$reviews.rating" },
      totalReviews: { $sum: 1 },
      author: { $first: "$author" }
    }
  },
  { $sort: { avgReviewRating: -1 } },
  {
    $project: {
      _id: 0,
      title: "$_id",
      author: 1,
      avgReviewRating: { $round: ["$avgReviewRating", 2] },
      totalReviews: 1
    }
  }
])
< {
  totalReviews: 2,
  author: 'R. Menon',
  title: 'The Silent Algorithm',
  avgReviewRating: 4.5
}

```

Task 4: Indexing for Speed

- Created a text index on reviews.comment.
- Used \$text to search for the word Digital.
- Used .explain("executionStats") to check index usage.
- TEXT_MATCH and IXSCAN show that the text index was used.
- Only 1 document was examined and returned in this query.
- Indexing improves search performance by avoiding a full collection scan.

```

> db.book_metadata.createIndex({
  "reviews.comment": "text"
})
< reviews.comment_text
> db.book_metadata.find(
  { $text: { $search: "Digital" } },
  {
    title: 1,
    score: { $meta: "textScore" },
    _id: 0
  }
).sort({
  score: { $meta: "textScore" }
})
< {
  title: 'The Silent Algorithm',
  score: 1.1833333333333333
}
> db.book_metadata.find(
  { $text: { $search: "Digital" } }
).explain("executionStats")

```

```

< {
  explainVersion: '1',
  queryPlanner: {
    namespace: 'BookFlowDB.book_metadata',
    parsedQuery: {
      '$text': {
        '$search': 'Digital',
        '$language': 'english',
        '$caseSensitive': false,
        '$diacriticSensitive': false
      }
    }
  },

```

```
indexFilterSet: false,
queryHash: '4FB85230',
planCacheShapeHash: '4FB85230',
planCacheKey: 'F34F3876',
optimizationTimeMillis: 1,
maxIndexedOrSolutionsReached: false,
maxIndexedAndSolutionsReached: false,
maxScansToExplodeReached: false,
prunedSimilarIndexes: false,
winningPlan: {
  isCached: false,
  stage: 'TEXT_MATCH',
  nss: 'BookFlowDB.book_metadata',
  indexPrefix: {},
  indexName: 'reviews.comment_text',
  parsedTextQuery: {
    terms: [
      'digit'
    ],
    negatedTerms: [],
    phrases: [],
    negatedPhrases: []
  },
},
```

```
textIndexVersion: 3,
inputStage: {
  stage: 'FETCH',
  nss: 'BookFlowDB.book_metadata',
  inputStage: {
    stage: 'IXSCAN',
    nss: 'BookFlowDB.book_metadata',
    keyPattern: {
      _fts: 'text',
      _ftsx: 1
    },
    indexName: 'reviews.comment_text',
    isMultiKey: true,
    isUnique: false,
    isSparse: false,
    isPartial: false,
    indexVersion: 2,
    direction: 'backward',
    indexBounds: {}
  }
},
rejectedPlans: []
},
```

```
executionStats: {
  executionSuccess: true,
  nReturned: 1,
  executionTimeMillis: 6,
  totalKeysExamined: 1,
  totalDocsExamined: 1,
  executionStages: {
    isCached: false,
    stage: 'TEXT_MATCH',
    nReturned: 1,
    executionTimeMillisEstimate: 0,
    works: 2,
    advanced: 1,
    needTime: 0,
    needYield: 0,
    saveState: 0,
    restoreState: 0,
    isEOF: 1,
    nss: 'BookFlowDB.book_metadata',
    indexPrefix: {},
    indexName: 'reviews.comment_text',
    parsedTextQuery: {
      terms: [
        'digit'
      ],
      negatedTerms: [],
      phrases: [],
      negatedPhrases: []
    }
  }
},
```



```
textIndexVersion: 3,
docsRejected: 0,
inputStage: {
  stage: 'FETCH',
  nReturned: 1,
  executionTimeMillisEstimate: 0,
  works: 2,
  advanced: 1,
  needTime: 0,
  needYield: 0,
  saveState: 0,
  restoreState: 0,
  isEOF: 1,
  nss: 'BookFlowDB.book_metadata',
  docsExamined: 1,
  alreadyHasObj: 0,
  inputStage: {
    stage: 'IXSCAN',
    nReturned: 1,
    executionTimeMillisEstimate: 0,
    works: 2,
    advanced: 1,
    needTime: 0,
    needYield: 0,
    saveState: 0,
    restoreState: 0,
    isEOF: 1,
    nss: 'BookFlowDB.book_metadata',
```

```
    keyPattern: {
      _fts: 'text',
      _ftsx: 1
    },
    indexName: 'reviews.comment_text',
    isMultiKey: true,
    isUnique: false,
    isSparse: false,
    isPartial: false,
    indexVersion: 2,
    direction: 'backward',
    indexBounds: {},
    keysExamined: 1,
    seeks: 1,
    dupsTested: 1,
    dupsDropped: 0,
    peakTrackedMemBytes: 8
  }
}
},
```

```
queryShapeHash: '8238985C7B3F1CE45E80D77A5F01FECBF0618DD749B1B6A41990FCF19C3B5FC'
peakTrackedMemBytes: Long('8'),
command: {
  find: 'book_metadata',
  filter: {
    '$text': {
      '$search': 'Digital'
    }
  },
  '$db': 'BookFlowDB'
},
serverInfo: {
  host: 'DESKTOP-BLT5L3R',
  port: 27017,
  version: '8.3.8',
  gitVersion: '35e8c8a57f78157ed9fac1a9e90ee6c1818adab6'
},
serverParameters: {
  internalQueryFacetBufferSizeBytes: 104857600,
  internalDocumentSourceGroupMaxMemoryBytes: 104857600,
  internalQueryMaxBlockingSortMemoryUsageBytes: 104857600,
  internalDocumentSourceSetWindowFieldsMaxMemoryBytes: 104857600,
  internalQueryFacetMaxOutputDocSizeBytes: 104857600,
  internalLookupStageIntermediateDocumentMaxSizeBytes: 104857600,
  internalQueryProhibitBlockingMergeOnMongoS: 0,
  internalQueryMaxAddToSetBytes: 104857600,
  internalQueryFrameworkControl: 'trySbeRestricted',
  internalQueryPlannerIgnoreIndexWithCollationForRegex: 1
}
```

```
    },
    ok: 1
  }
> db.book_metadata.find(
  { "reviews.comment": { $regex: "Digital", $options: "i" } }
).explain("executionStats")
< {
  explainVersion: '1',
  queryPlanner: {
    namespace: 'BookFlowDB.book_metadata',
    parsedQuery: {
      'reviews.comment': BSONRegExp('Digital', 'i')
    },
    indexFilterSet: false,
    queryHash: '819E6067',
    planCacheShapeHash: '819E6067',
    planCacheKey: '29FFD58A',
    optimizationTimeMillis: 0,
    maxIndexedOrSolutionsReached: false,
    maxIndexedAndSolutionsReached: false,
    maxScansToExplodeReached: false,
    prunedSimilarIndexes: false,
    winningPlan: {
      isCached: false,
      stage: 'COLLSCAN',
      filter: {
        'reviews.comment': BSONRegExp('Digital', 'i')
      },
    },
  },
}
```

```
    nss: 'BookFlowDB.book_metadata',
    direction: 'forward'
  },
  rejectedPlans: []
},
executionStats: {
  executionSuccess: true,
  nReturned: 1,
  executionTimeMillis: 0,
  totalKeysExamined: 0,
  totalDocsExamined: 1,
  executionStages: {
    isCached: false,
    stage: 'COLLSCAN',
    filter: {
      'reviews.comment': BSONRegExp('Digital', 'i')
    },
    nReturned: 1,
    executionTimeMillisEstimate: 0,
    works: 2,
    advanced: 1,
    needTime: 0,
    needYield: 0,
    saveState: 0,
    restoreState: 0,
    isEOF: 1,
    nss: 'BookFlowDB.book_metadata',
    direction: 'forward',
```

```
    }  
  },  
  queryShapeHash: 'B8A08CA9817196B46FC1ABCA53960DA69592D3082AB6BD4C547D266F035B054',  
  command: {  
    find: 'book_metadata',  
    filter: {  
      'reviews.comment': BSONRegExp('Digital', 'i')  
    },  
    '$db': 'BookFlowDB'  
  },  
  serverInfo: {  
    host: 'DESKTOP-BLT5L3R',  
    port: 27017,  
    version: '8.3.8',  
    gitVersion: '35e8c8a57f78157ed9fac1a9e90ee6c1818adab6'  
  },  
  serverParameters: {  
    internalQueryFacetBufferSizeBytes: 104857600,  
    internalDocumentSourceGroupMaxMemoryBytes: 104857600,  
    internalQueryMaxBlockingSortMemoryUsageBytes: 104857600,  
    internalDocumentSourceSetWindowFieldsMaxMemoryBytes: 104857600,  
    internalQueryFacetMaxOutputDocSizeBytes: 104857600,  
    internalLookupStageIntermediateDocumentMaxSizeBytes: 104857600,  
    internalQueryProhibitBlockingMergeOnMongoS: 0,  
    internalQueryMaxAddToSetBytes: 104857600,  
    internalQueryFrameworkControl: 'trySbeRestricted',  
    internalQueryPlannerIgnoreIndexWithCollationForRegex: 1  
  },  
}
```